EE5334 Assignment 1

Objective 1: Vowel and Consonant Analysis of Telugu Speech

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Abstract

Telugu vowel and consonant recordings were analysed using time-domain and frequency-domain methods. Three voiced vowel tokens, Amma (/a/), Illu (/i/), and Uduta (/u/), were examined using narrowband spectrograms, harmonic spacing, the average magnitude difference function (AMDF), and wideband spectrograms with LPC-based formant tracks. The harmonic-spacing and AMDF pitch estimates differed by 1.11 Hz for Amma, 1.61 Hz for Illu, and 1.74 Hz for Uduta.

The consonant recording contained 25 tokens arranged in five place-of-articulation groups: velar, palatal, retroflex, apico-dental, and labial. Individual waveforms and short-time spectrograms were examined for closure intervals, release bursts, aspiration noise, voicing, nasal resonance, and transitions into the following vowel. The observations show how manner-related cues appear in the time-frequency representation, while place distinctions are supported by burst spectra and formant transitions.

1 Objective

The objective is to record and observe vowel and consonant examples from an Indian language and to analyse selected voiced sounds. The vowel-analysis requirements are: (i) narrowband spectrograms for observing harmonic structure and deducing pitch, (ii) pitch estimation using the AMDF or a related method, and (iii) wideband spectrograms on which the first three formant contours are marked. The consonant recording is organised according to the place- and manner-of-articulation chart supplied with the assignment. Telugu was selected as the language for the recordings.

2 Speech Material and Recording

2.1 Recording Format

The recordings were prepared using the sampling frequency and bit depth specified in the assignment. Their properties are summarised in Table 1.

Table 1: Recording properties.

Parameter	Recorded value
Sampling frequency	44.1 kHz
Bit depth	16 bits/sample
Channels	Mono
Vowel recording duration	Approximately 22.39 s
Consonant recording duration	Approximately 55.93 s

The WAV files were read in MATLAB using `audioread`. The vowel signal was resampled to 10 kHz for pitch and LPC processing. A pre-emphasis filter with coefficient 0.8 was applied before LPC and cepstral processing. The consonant spectrograms were calculated at the original 44.1 kHz sampling rate to retain the high-frequency release and frication components.

2.2 Vowel Material and Selected Segments

Twelve Telugu word tokens were recorded to represent the vowel examples listed in Table 2. The current detailed vowel analysis uses Amma, Illu, and Uduta. Their approximate boundaries are listed in Table 3.

Table 2: Recorded Telugu vowel examples.

No.	Word	Vowel	No.	Word	Vowel
1	Amma	/a/	7	Eluka	/e/
2	Aavu	/aa/	8	Eenugu	/ee/
3	Illu	/i/	9	Aidu	/ai/
4	Eega	/ii/	10	Onte	/o/
5	Uduta	/u/	11	Oda	/oo/
6	Ooyala	/uu/	12	Aushadham	/au/

Table 3: Approximate word and target-vowel boundaries for the three analysed tokens.

Word	Word interval (s)	Vowel interval (s)	Target vowel
Amma	1.74–2.44	1.77–1.94	/a/
Illu	4.76–5.47	4.83–5.00	/i/
Uduta	8.19–8.88	8.22–8.39	/u/

The boundaries are approximate segmentation times. The vowel portions were isolated for pitch estimation, while a short amount of surrounding speech was retained in the spectrograms to show transitions into and out of each vowel.

3 Consonant Classification

3.1 Place and Manner of Articulation

The consonant samples follow the five-row and five-column arrangement in the assignment chart. The rows represent place of articulation, and the columns represent voiceless unaspirated, voiceless aspirated, voiced unaspirated, voiced aspirated, and nasal manners. The transliterations in Table 4 are retained to match the recording labels and the assignment chart.

Table 4: Consonant set arranged by place and manner of articulation.

Place	Voiceless unaspirated	Voiceless aspirated	Voiced unaspirated	Voiced aspirated	Nasal
Velar	ka	kha	ga	gha	nga
Palatal series	ca	cha	ja	jha	nya
Retroflex	Ta	Tha	Da	Dha	Na
Apico-dental	ta	tha	da	dha	na
Labial	pa	pha	ba	bha	ma

Capital letters `T`, `D`, and `N` identify the retroflex series and distinguish it from the lower-case apico-dental series. The place labels follow the assignment chart. In detailed descriptions of Telugu, some of the relevant places are more narrowly distinguished as dentialveolar and palato-alveolar; the palatal-series affricates are not identical in articulation to a palatal approximant [2].

- **Velar:** The main constriction is made between the back of the tongue and the soft palate. The corresponding nasal is the velar nasal /ng/, represented here as `nga`.
- **Palatal series:** The tongue body and the region behind the alveolar ridge form the constriction. The Telugu series represented by `ca, cha, ja, jha` is commonly realised with an affricated release; `nya` is the corresponding palatal nasal.
- **Retroflex:** The tongue tip is retracted or curled back toward the postalveolar region. The retroflex series contrasts with the apico-dental series in tongue position.
- **Apico-dental:** The tongue tip or blade approaches the upper teeth or the immediately adjacent dental region. This is the label used for the corresponding row in the assignment chart.
- **Labial:** The constriction is formed by the lips. The nasal `ma` is produced with lip closure and a lowered velum.

3.2 Voicing and Aspiration

The stop columns distinguish two related but separate properties. Voicing refers to vocal-fold vibration, while aspiration refers to the interval of turbulent airflow following the oral release.

These properties are examined using the release burst, the onset of periodic energy, and the interval between them.

Voice-onset time (VOT) is defined as

$$\mathrm{VOT} = t_{\text{voicing onset}} - t_{\text{release}}. \tag{1}$$

For a voiceless unaspirated stop, voicing begins near the release; a voiceless aspirated stop generally has a longer positive VOT. In a voiced stop, low-frequency periodic energy may begin before or close to the oral release. The VOT distinction is a timing cue and should not be inferred from burst amplitude alone [3, 4].

Nasals are produced with an oral closure at the relevant place and an open velopharyngeal passage. They are normally voiced and show resonances and anti-resonances associated with the coupled oral and nasal cavities. Their spectral envelope is generally weaker and more irregular than that of an adjacent vowel [3].

3.3 Telugu-Specific Phonetic Considerations

The traditional chart is useful for demonstrating the place and manner contrasts in the assignment, but it should not be interpreted as meaning that all aspirated categories occur equally often in native Telugu vocabulary. Bhaskararao and Ray report that aspirated consonants occur mainly in Tatsama and Tadbhava words derived from Sanskrit and Prakrit; several aspirated consonants are rare in native vocabulary. They also describe place distinctions such as dentialveolar and palato-alveolar articulations in the Telugu inventory [2].

The recorded set is therefore treated as a controlled recitation of the categories in the assignment chart rather than as a frequency survey of consonants in everyday Telugu words.

4 Consonant Recording and Segmentation

4.1 Recording Order

The consonant recording contains 25 labelled CV tokens in five successive groups of five. The sequence follows the rows of the assignment chart: velar, palatal series, retroflex, apico-dental, and labial. The groups are separated by low-energy intervals in the full-recording waveform. This ordering makes within-place comparisons possible: for example, `ka` and `kha` share the same place label but differ in aspiration, while `ga` and `gha` share voicing but differ in aspiration.

4.2 Token Intervals

The approximate intervals used to extract the consonant tokens are given in Table 5. These are analysis windows that include the consonant release and the following vowel transition; they are not claimed to be exact articulatory closure boundaries.

Table 5: Approximate extraction intervals for the 25 consonant tokens.

No.	Token	Place	Manner	Start (s)	End (s)
1	ka	Velar	Voiceless unaspirated	2.55	3.20
2	kha	Velar	Voiceless aspirated	4.16	4.99
3	ga	Velar	Voiced unaspirated	6.00	6.71
4	gha	Velar	Voiced aspirated	8.02	8.87
5	nga	Velar	Nasal	9.75	10.60

No.	Token	Place	Manner	Start (s)	End (s)
6	ca	Palatal series	Voiceless unaspirated	12.28	12.97
7	cha	Palatal series	Voiceless aspirated	14.48	15.24
8	ja	Palatal series	Voiced unaspirated	16.41	17.23
9	jha	Palatal series	Voiced aspirated	18.67	19.63
10	nya	Palatal series	Nasal	20.90	21.88
11	Ta	Retroflex	Voiceless unaspirated	23.70	24.51
12	Tha	Retroflex	Voiceless aspirated	25.73	26.53
13	Da	Retroflex	Voiced unaspirated	27.76	28.63
14	Dha	Retroflex	Voiced aspirated	30.25	31.21
15	Na	Retroflex	Nasal	32.24	33.16
16	ta	Apico-dental	Voiceless unaspirated	34.25	34.92
17	tha	Apico-dental	Voiceless aspirated	36.32	37.03
18	da	Apico-dental	Voiced unaspirated	38.35	39.32
19	dha	Apico-dental	Voiced aspirated	40.96	41.78
20	na	Apico-dental	Nasal	43.26	44.10
21	pa	Labial	Voiceless unaspirated	45.58	46.43
22	pha	Labial	Voiceless aspirated	47.71	48.49
23	ba	Labial	Voiced unaspirated	49.61	50.43
24	bha	Labial	Voiced aspirated	51.68	52.55
25	ma	Labial	Nasal	53.74	54.73

4.3 Consonant Spectrogram Parameters

The consonant signal was analysed at its original 44.1 kHz sampling rate. A 5 ms Hamming window and a 1 ms frame shift were used to obtain time-resolved spectrograms. The FFT length was 4096, and plots were displayed up to 12 kHz. Each token plot contains its time-domain waveform and short-time spectrogram. The complete recording waveform and spectrogram were also generated to show the order and separation of the five groups.

5 Acoustic Principles Used for Consonant Observation

5.1 Stops and Release Bursts

A stop consonant typically contains a closure interval, a release transient, and a transition into the following vowel. In a spectrogram, closure appears as a reduction in energy, while the release burst appears as a brief vertical concentration of energy. The following vowel produces periodic harmonics and formant bands.

The burst and the formant transitions provide information about place of articulation; the closure, voicing, and aspiration intervals provide information about manner [3].

5.2 Aspiration and Voicing

Aspiration is not the same event as the release burst. The burst is brief, whereas aspiration appears as a longer interval of aperiodic noise after release and before regular vowel voicing. A longer delay between release and voicing is consistent with greater positive VOT. Voiced stops can show low-frequency periodic energy around or during the closure, while unaspirated voiceless stops generally show a shorter release-to-voicing interval than their aspirated counterparts [3, 4].

5.3 Nasals

Nasals are distinguished from oral stops by the acoustic effect of nasal-cavity coupling. A nasal often contains continuous low-frequency periodic energy, weaker upper-frequency energy, and spectral valleys associated with anti-formants.

The transition between the nasal and the following vowel is also informative because the oral closure is released while the nasal passage remains open [3].

5.4 Place of Articulation

Place is not identified reliably from one spectral peak alone. It is inferred from a combination of release-burst spectral shape and the trajectories of the first formants in the following vowel. Bilabial, dental/alveolar, and velar releases can differ in their spectral concentration, while retroflex and dental consonants can produce different transitions into the same vowel.

These cues depend on the speaker, vowel context, and recording conditions; reference frequency ranges are therefore not used as fixed thresholds for the Telugu tokens [3].

6 Consonant Observations from the Recording

6.1 Overall Recording Structure

The complete waveform shows 25 high-energy utterance regions arranged in five successive groups. Low-energy gaps separate the individual tokens and make the five-place sequence visible in the time-domain recording.

The corresponding full-recording spectrogram shows that each utterance contains a consonantal onset followed by voiced energy from the accompanying vowel. The within-token spectrograms are therefore interpreted as CV sequences: the consonant release or nasal portion is examined together with the transition into the vowel.

6.2 Observations by Place Group

1. **Velar group (ka, kha, ga, gha, nga):** The four oral-stop tokens show a transient onset followed by periodic vowel energy. In the aspirated-labelled examples, the release region contains a more extended diffuse high-frequency component than a simple short burst. The voiced examples show low-frequency periodic energy closer to the vowel onset.
   The nasal `nga` has a more continuous voiced structure and does not show the same abrupt closure-release pattern as the oral stops. The spectral shape of the release and the transition into the vowel are the main cues used to distinguish the velar group from the other places.
2. **Palatal series (ca, cha, ja, jha, nya):** The stop/affricate tokens show concentrated release energy followed by a noisy component and vowel formants. The noisy part is more extended in the aspirated-labelled tokens. This is consistent with the affricated release described for the corresponding Telugu palato-alveolar series.
   The nasal `nya` instead shows sustained voiced energy and a less abrupt release pattern. The high-frequency portion of the release and the following formant transitions are useful for comparing this group with the velar and retroflex groups.
3. **Retroflex group (Ta, Tha, Da, Dha, Na):** The four oral-stop tokens show closure-to-release transitions followed by vowel periodicity. The aspiration-labelled tokens contain a more visible aperiodic interval after release than the unaspirated counterpart. The voiced-labelled tokens show low-frequency energy nearer to the release-vowel transition.
   The nasal `Na` has a sustained resonant structure rather than a strong isolated plosive burst. The retroflex-versus-dental distinction is assessed from the release spectrum and the direction of

formant transitions into the following vowel, rather than from amplitude alone.

4. **Apico-dental group (ta, tha, da, dha, na):** The stop tokens contain brief release events followed by vowel formant bands. The aspirated-labelled examples show a longer noise-bearing interval after release than the voiceless unaspirated example. The voiced tokens contain periodic low-frequency energy around the onset of voicing.
   The nasal `na` shows a comparatively continuous voiced portion. Differences from the retroflex group are most usefully examined in the spectral distribution of the release and in the vowel transitions; the place decision is not made from a single frequency bin.
5. **Labial group (pa, pha, ba, bha, ma):** The oral-stop tokens show closure and release patterns followed by voiced vowel energy. The aspirated examples contain additional aperiodic energy after the release, while the voiced examples show earlier low-frequency periodicity.
   The nasal `ma` has a sustained voiced resonance pattern and a less explosive onset than the oral stops. Burst-spectrum shape and the transitions into the vowel provide the main place cues for comparison with the dental and velar groups.

6.3 Cross-Group Comparison

The within-place comparisons are particularly useful because the place label remains fixed while voicing or aspiration changes. For example, `ka-kha` compares unaspirated and aspirated voiceless velars, while `ga-gha` compares the corresponding voiced pair. The same comparison is repeated for the palatal, retroflex, apico-dental, and labial groups.

Across these pairs, the main acoustic distinction is the duration and strength of the post-release noise and the timing of periodic voicing. The five nasal tokens provide a separate comparison across place while keeping nasal manner constant.

The spectrograms also show why waveform amplitude alone is insufficient for classification. A release burst is short and may occupy only a few milliseconds; aspiration is identified by its following noise interval; voicing is identified by periodic low-frequency structure; and place is inferred from burst spectrum together with vowel transitions.

These cues are consistent with standard acoustic descriptions of stops, affricates, and nasals [3, 4].

6.4 Limitations of the Consonant Measurements

The intervals in Table 5 are broad extraction windows centred on the recorded tokens. They include the consonant and the following vowel transition and are not exact phone-boundary annotations.

The present analysis records qualitative observations from waveform and spectrogram structure; numerical VOT values, burst spectral-centroid values, and nasal formant frequencies were not measured. Such quantities would require manually verified release and voicing landmarks and a consistent measurement procedure across all tokens.

The classification is therefore based on the recorded sequence, the visible acoustic cues, and the articulation categories in the assignment chart.

7 Vowel Analysis Method

7.1 Analysis Settings

The main vowel-processing parameters are listed in Table 6.

Table 6: Task 1 vowel-processing parameters.

Parameter	Setting
Original sampling rate	44.1 kHz
Analysis sampling rate for pitch and LPC	10 kHz
Pre-emphasis coefficient	0.8
Pitch frame duration and shift	40 ms and 10 ms
Pitch search range	60–350 Hz
Narrowband window and shift	40 ms Hamming window; 10 ms shift
Narrowband FFT length	8192
Averaged harmonic-spectrum frames	40 ms duration; 10 ms shift
Harmonic-spectrum FFT length	32768
Wideband window and shift	5 ms Hamming window; 1 ms shift
Wideband FFT length	2048
LPC frame duration and shift	25 ms and 5 ms
LPC order	14
Displayed wideband frequency range	0–4000 Hz

7.2 Narrowband Spectrogram and Harmonic Pitch

A relatively long analysis window gives finer frequency resolution and reveals the harmonic structure of voiced speech. The harmonics occur approximately at

$$f_h = hf_0, \qquad h = 1, 2, 3, \ldots, \tag{2}$$

where f_0 is the fundamental frequency. The difference between consecutive harmonics is approximately f_0. The harmonic-spacing estimate uses an averaged magnitude spectrum and checks whether detected peak gaps are close to integer multiples of a candidate fundamental frequency. This reduces errors when a weak harmonic is not detected.

7.3 AMDF Pitch Estimation

The average magnitude difference function compares a signal with a delayed copy. For delay η, it is defined as

$$\mathrm{AMDF}(\eta) = \frac{1}{N-\eta} \sum_{n=0}^{N-\eta-1} |x(n+\eta) - x(n)|. \tag{3}$$

A delay close to the fundamental period produces a small difference because the waveform is approximately periodic. The period and pitch are related by

$$T_0 = \frac{\eta}{F_s}, \qquad f_0 = \frac{F_s}{\eta}. \tag{4}$$

The implementation evaluates overlapping frames, normalises the AMDF, checks candidate minima with correlation, and refines the selected delay by parabolic interpolation.

7.4 Wideband Spectrogram and Formant Estimation

A short analysis window gives improved time resolution for displaying the changing resonance bands of speech. The first three formants are estimated from LPC poles. For each frame, the LPC

polynomial is calculated, its roots are found, and the poles are converted to resonance-frequency and bandwidth candidates.

Candidates are filtered using pole radius, frequency, and bandwidth. Ordered F1, F2, and F3 candidates are selected using broad vowel-dependent ranges and continuity between frames. The selected tracks are overlaid on the wideband spectrogram.

The formant estimates are algorithmic resonance candidates rather than direct physical measurements. In particular, F3 is often weaker and more sensitive to model order, spectral-envelope shape, and competing poles. The wideband plots are therefore used to check whether the selected contour follows a visible resonance band.

8 Vowel Results

8.1 Measured Pitch and Formant Values

The measured pitch and formant estimates are summarised in Table 7.

Table 7: Task 1 pitch and formant estimates for the three selected vowels. All frequencies are in hertz.

Word / vowel	Harmonic pitch	AMDF pitch	F_1	F_2	F_3
Amma (/a/)	160.00	158.89	787.23	1151.33	3624.55
Illu (/i/)	150.75	152.36	321.15	2372.54	3636.10
Uduta (/u/)	148.75	150.49	413.87	857.62	2306.79

8.2 Agreement Between Pitch Methods

The absolute differences between harmonic-spacing and AMDF estimates are shown in Table 8.

Table 8: Difference between the two Task 1 pitch estimates.

Vowel	Harmonic pitch (Hz)	AMDF pitch (Hz)	Absolute difference (Hz)
Amma (/a/)	160.00	158.89	1.11
Illu (/i/)	150.75	152.36	1.61
Uduta (/u/)	148.75	150.49	1.74

The two methods give closely spaced estimates for all three vowels. The largest difference is 1.74 Hz for Uduta, while the smallest is 1.11 Hz for Amma. The estimates place the fundamental frequency close to 150–160 Hz for the three selected tokens.

8.3 Formant Observations

The measured formants show clear differences among the three vowel categories. Amma has the highest F1 at 787.23 Hz. Illu has the lowest F1 at 321.15 Hz and the highest F2 at 2372.54 Hz. Uduta has an intermediate F1 of 413.87 Hz and the lowest F2 at 857.62 Hz.

This ordering is consistent with the broad acoustic distinction between /a/, a relatively open vowel; /i/, a high front vowel; and /u/, a high back rounded vowel.

The F3 estimates for Amma and Illu are similar, at 3624.55 Hz and 3636.10 Hz. Uduta has a lower F3 estimate of 2306.79 Hz. Since F3 is often weaker and more sensitive to the LPC model, its value is interpreted together with the wideband contour rather than as an isolated number.

8.4 Observations During Vowel Analysis

The narrowband plots show regularly spaced harmonic components in all three voiced examples. The two pitch methods also give closely matching results, providing an independent check on the fundamental- frequency estimates.

The wideband plots and LPC tracks show the expected broad distinction between the low-F1/high-F2 pattern of /i/ and the lower-F2 pattern of /u/. The results also show that an LPC pole is only a resonance candidate: the candidate must be assessed from its bandwidth, position relative to the other formants, continuity across frames, and alignment with a visible resonance band.

9 Figures: Vowel Analysis

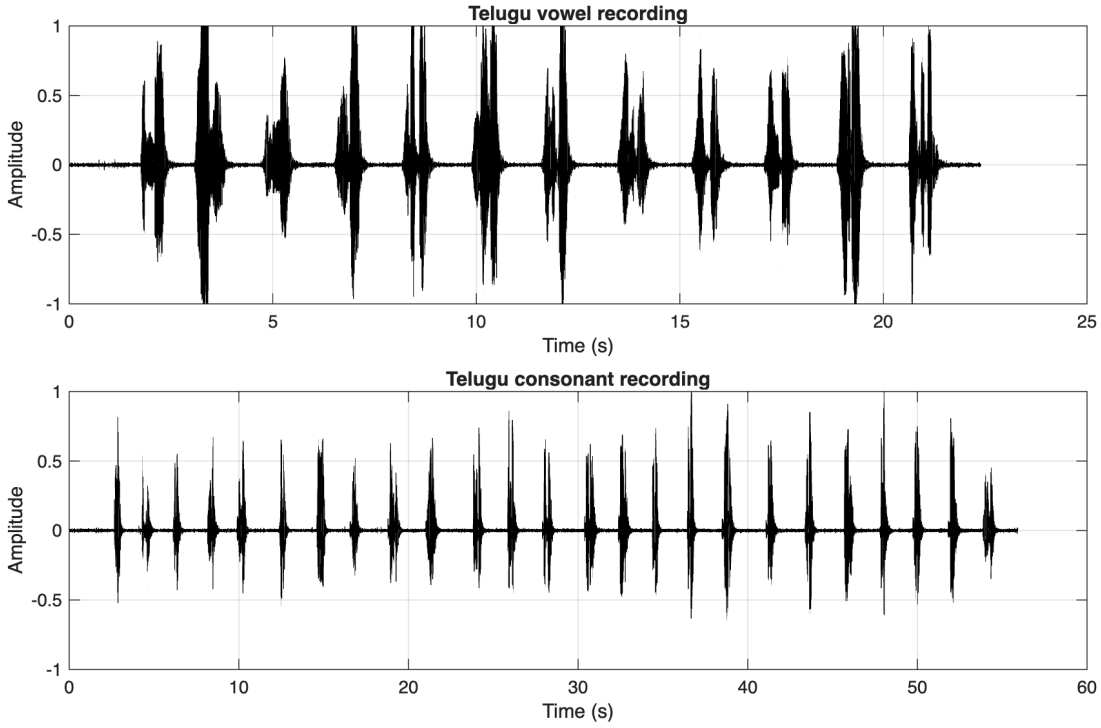
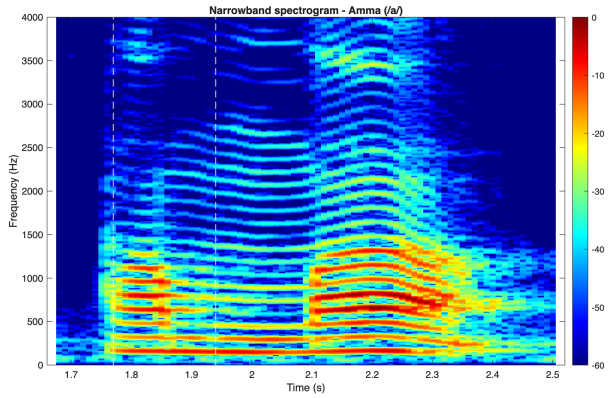
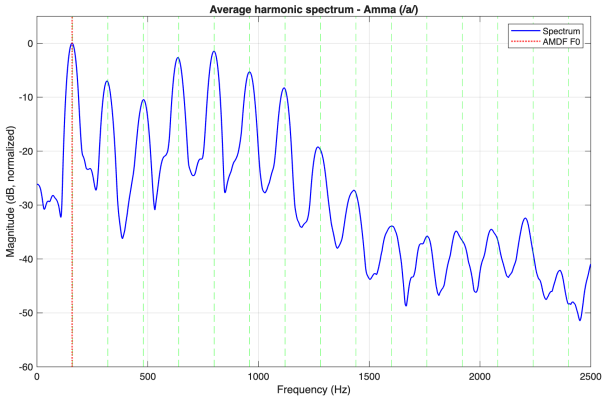


Figure 1: Overall vowel and consonant recording waveforms.

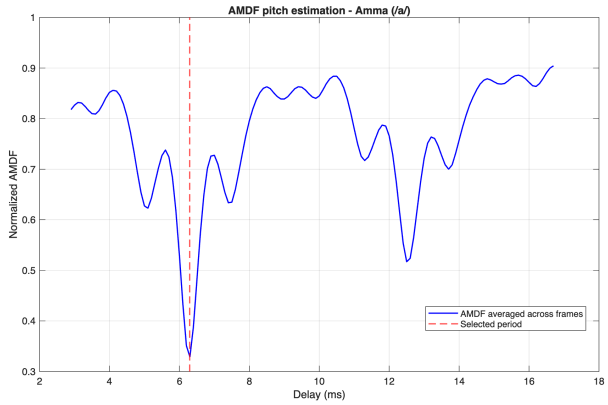
9.1 Amma (/a/)



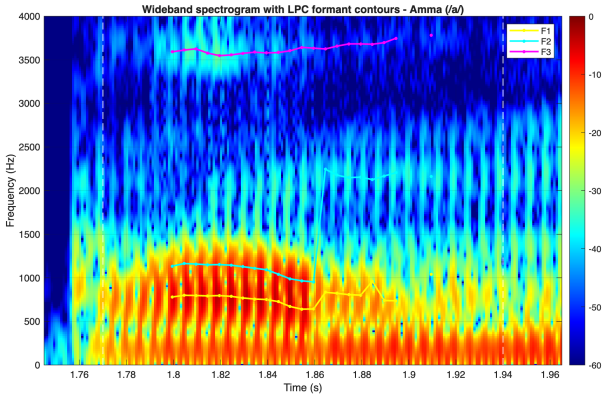
(a) Narrowband spectrogram.



(b) Harmonic spectrum.



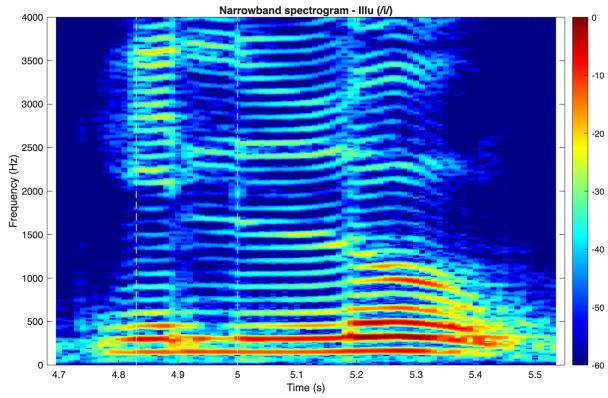
(c) AMDF curve.



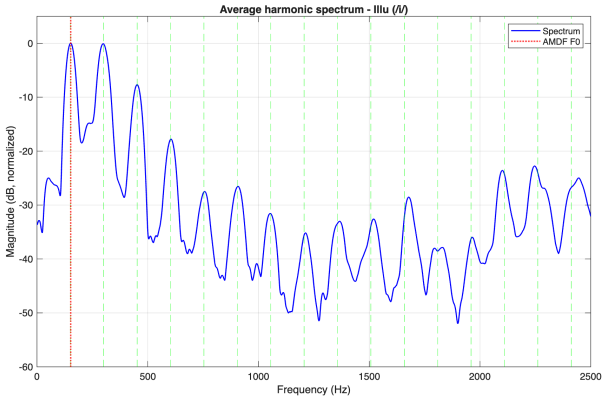
(d) Wideband spectrogram with formants.

Figure 2: Task 1 plots for Amma (/a/).

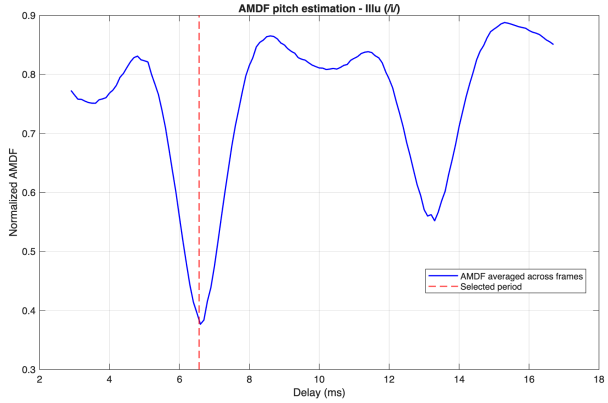
9.2 Illu (/i/)



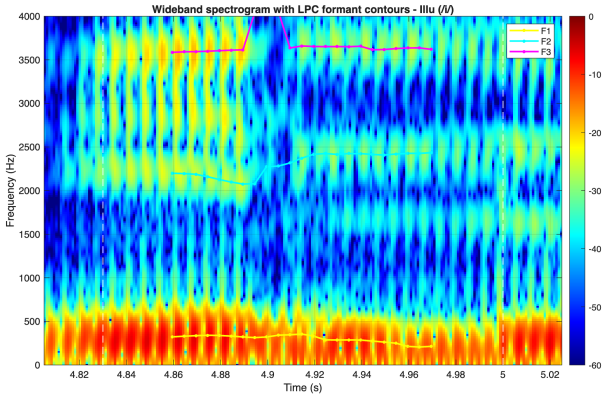
(a) Narrowband spectrogram.



(b) Harmonic spectrum.



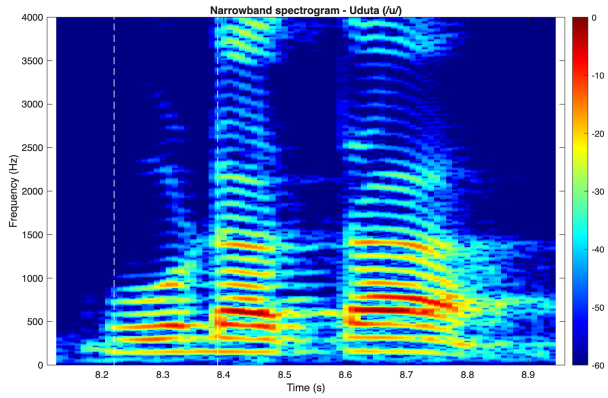
(c) AMDF curve.



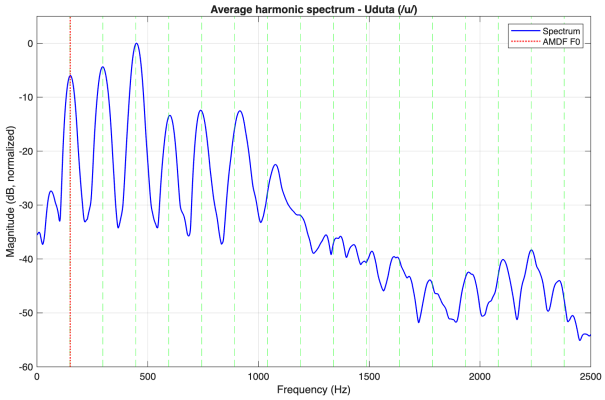
(d) Wideband spectrogram with formants.

Figure 3: Task 1 plots for Illu (/i/).

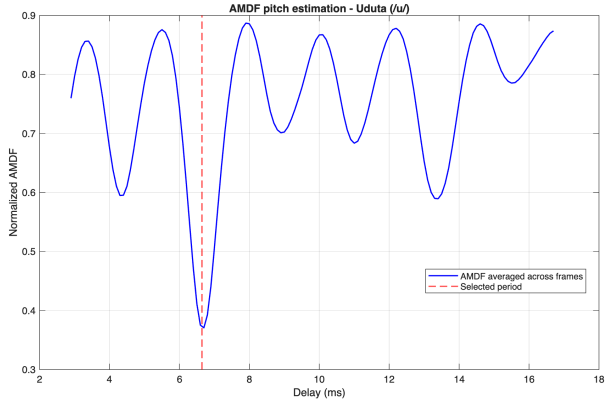
9.3 Uduta (/u/)



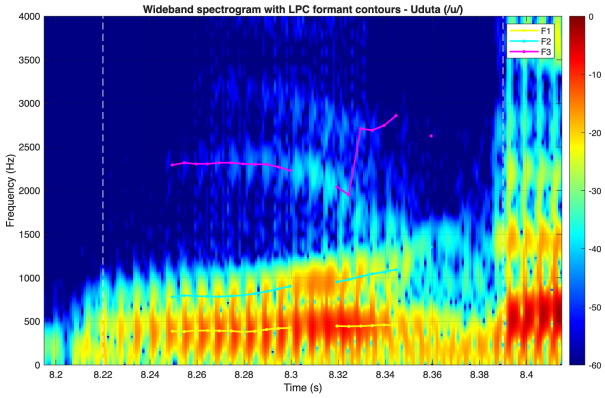
(a) Narrowband spectrogram.



(b) Harmonic spectrum.



(c) AMDF curve.



(d) Wideband spectrogram with formants.

Figure 4: Task 1 plots for Uduta (/u/).

10 Figures: Consonant Analysis

The following figures show the overall consonant recording and the individual waveform–spectrogram pairs for the 25 tokens. The token plots are grouped by place of articulation to allow comparisons of voicing and aspiration within each group.

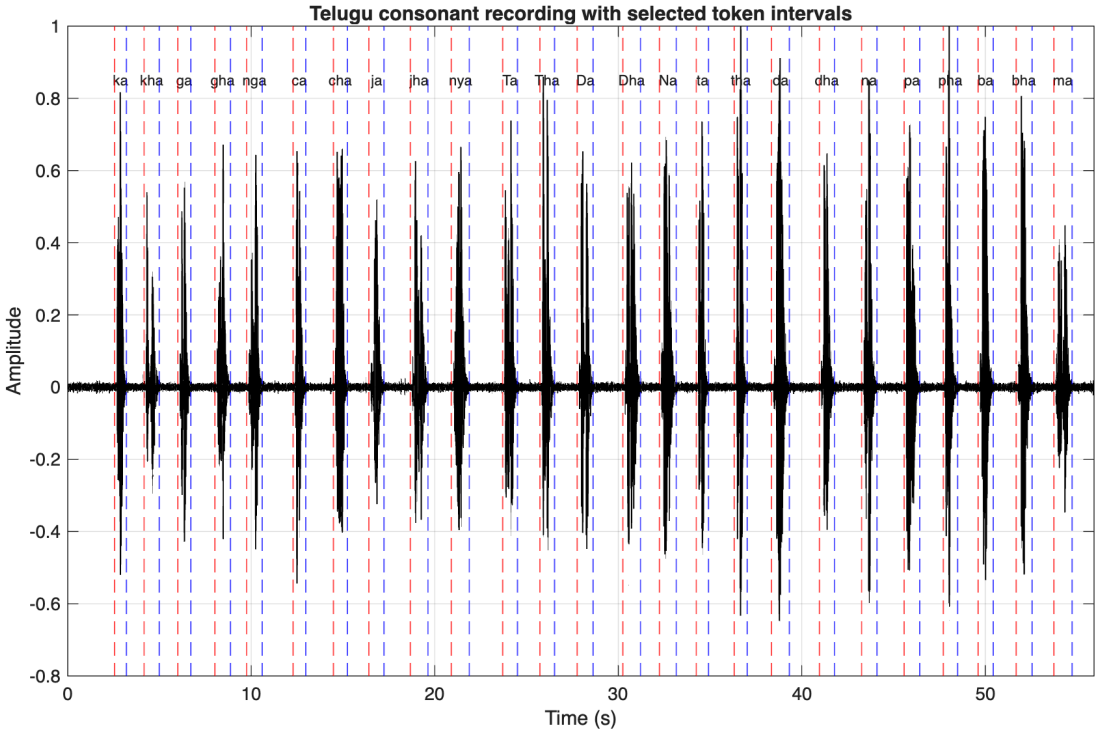


Figure 5: Complete consonant recording with the selected token intervals.

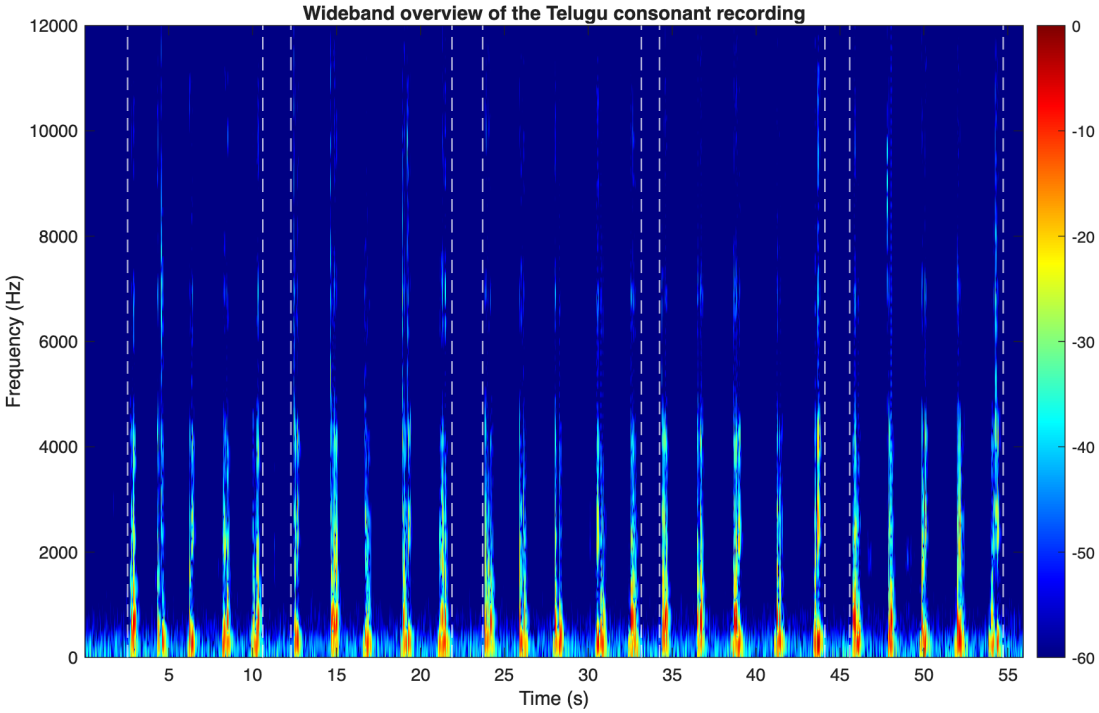
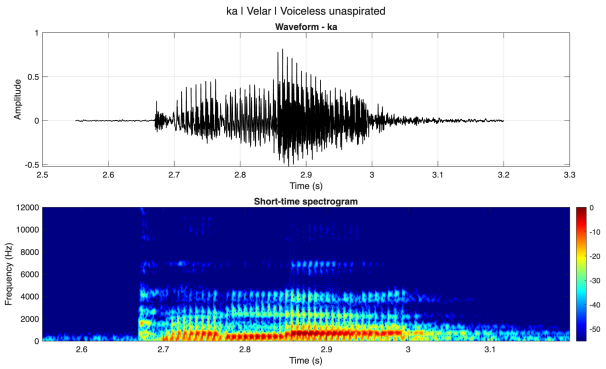
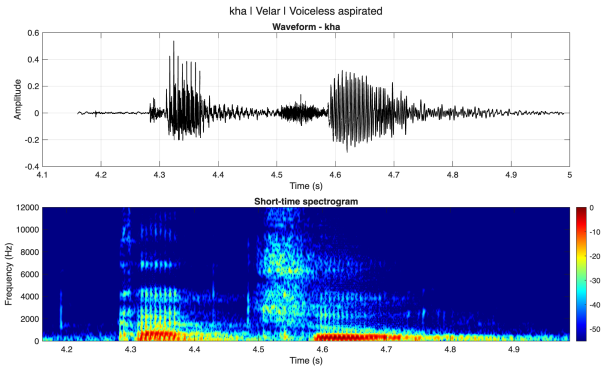


Figure 6: Wideband spectrogram of the complete consonant recording.

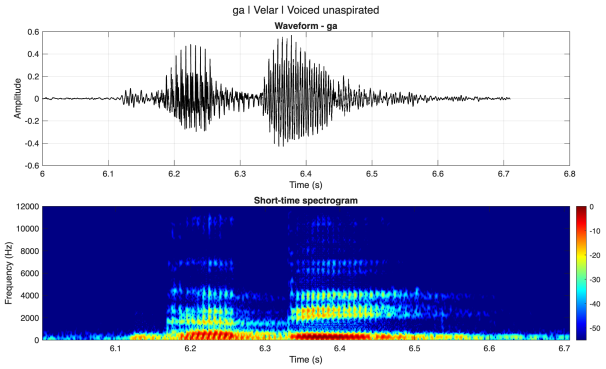
10.1 Velar Group



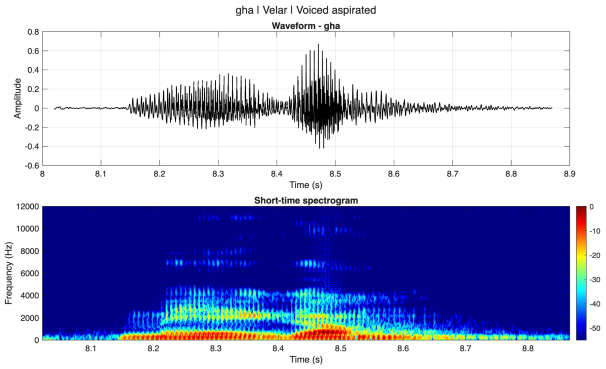
(a) ka: voiceless unaspirated.



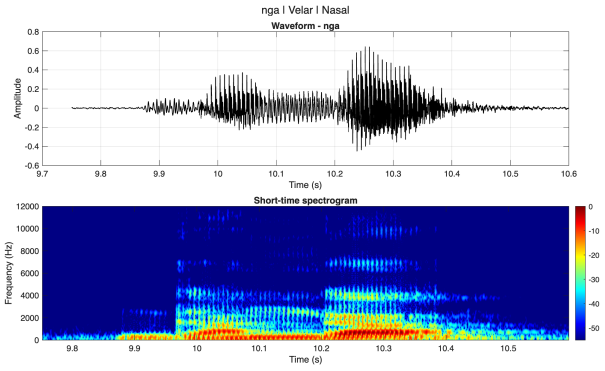
(b) kha: voiceless aspirated.



(c) ga: voiced unaspirated.



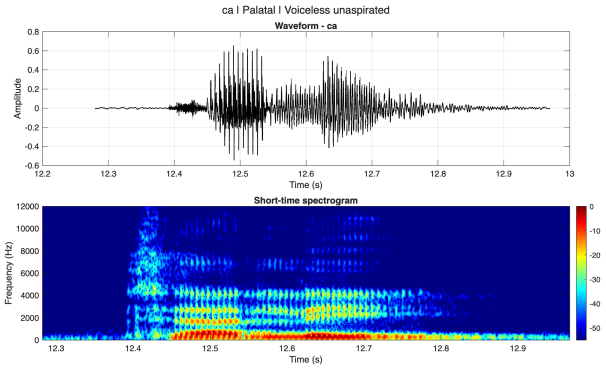
(d) gha: voiced aspirated.



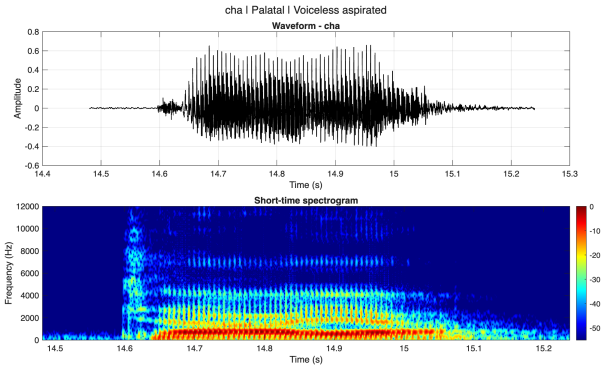
(e) nga: nasal.

Figure 7: Waveforms and spectrograms of the five velar tokens.

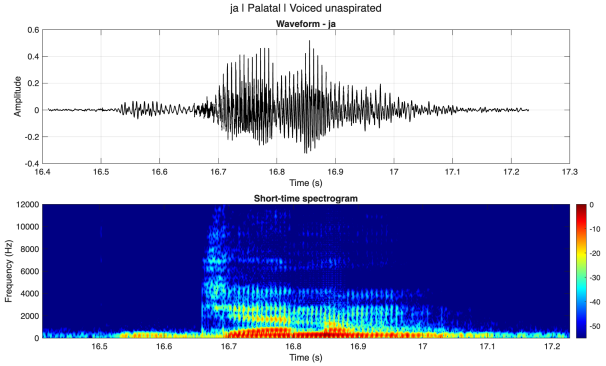
10.2 Palatal Series



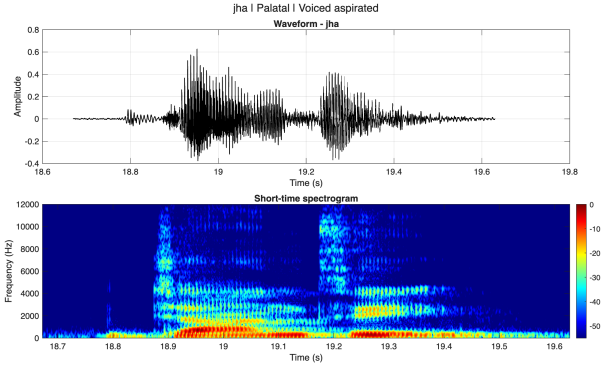
(a) ca: voiceless unaspirated.



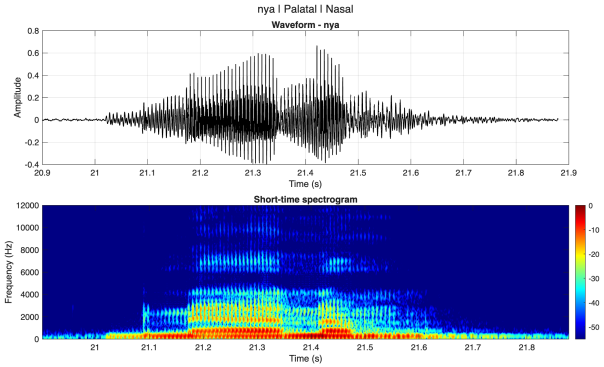
(b) cha: voiceless aspirated.



(c) ja: voiced unaspirated.



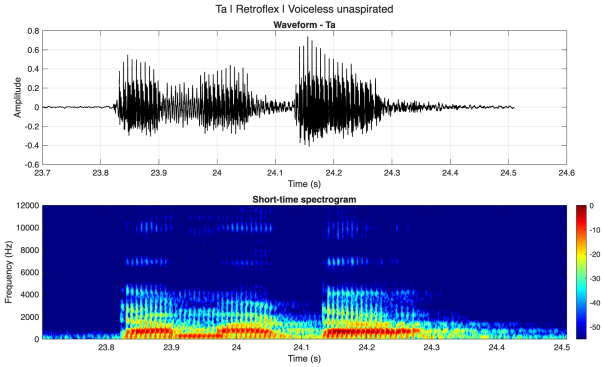
(d) jha: voiced aspirated.



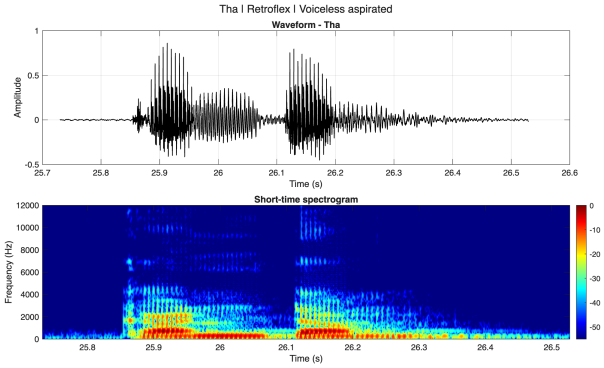
(e) nya: nasal.

Figure 8: Waveforms and spectrograms of the five palatal-series tokens.

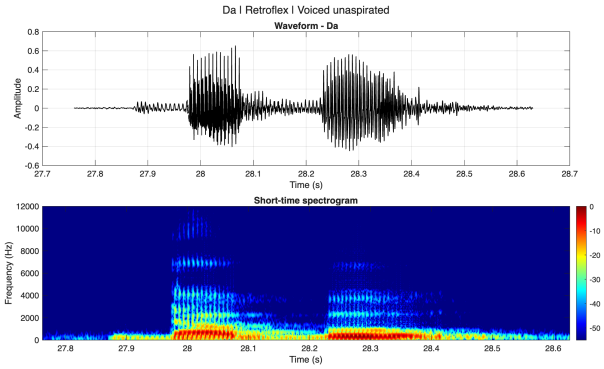
10.3 Retroflex Group



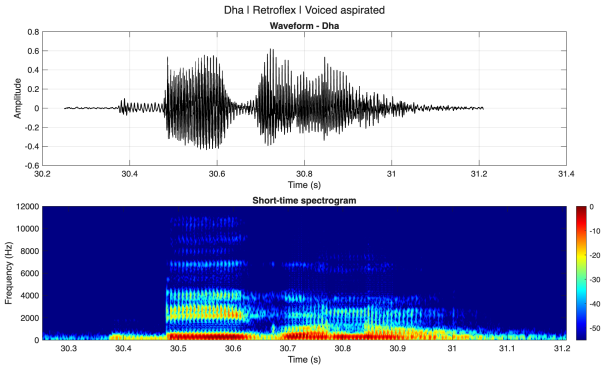
(a) Ta: voiceless unaspirated.



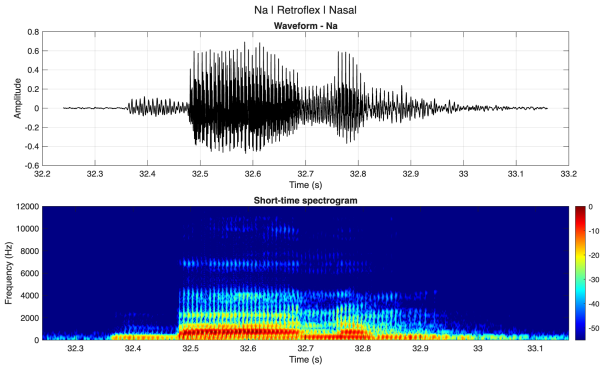
(b) Tha: voiceless aspirated.



(c) Da: voiced unaspirated.



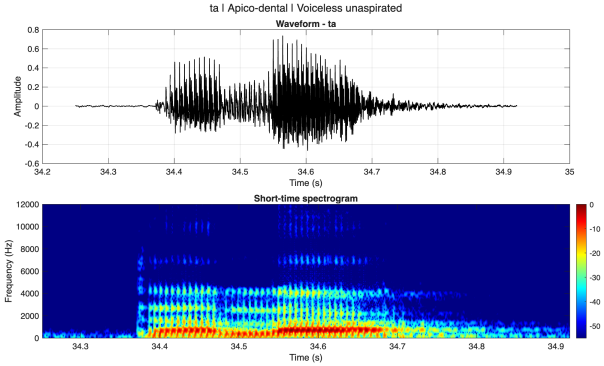
(d) Dha: voiced aspirated.



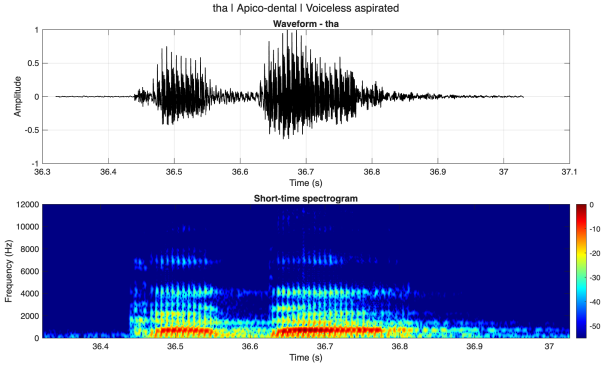
(e) Na: nasal.

Figure 9: Waveforms and spectrograms of the five retroflex tokens.

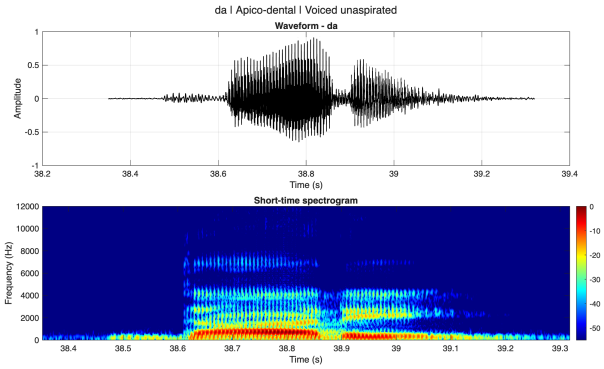
10.4 Apico-Dental Group



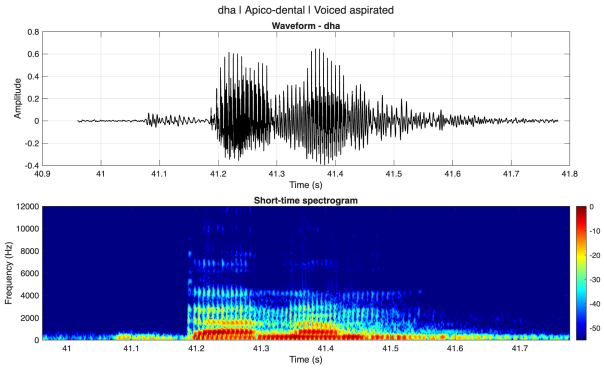
(a) ta: voiceless unaspirated.



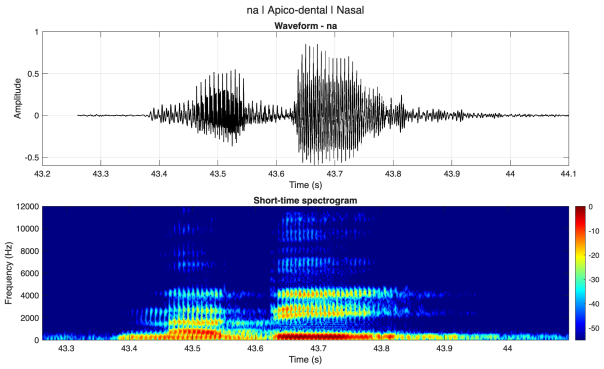
(b) tha: voiceless aspirated.



(c) da: voiced unaspirated.



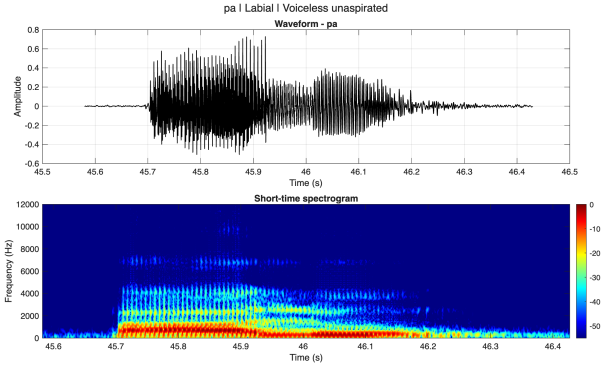
(d) dha: voiced aspirated.



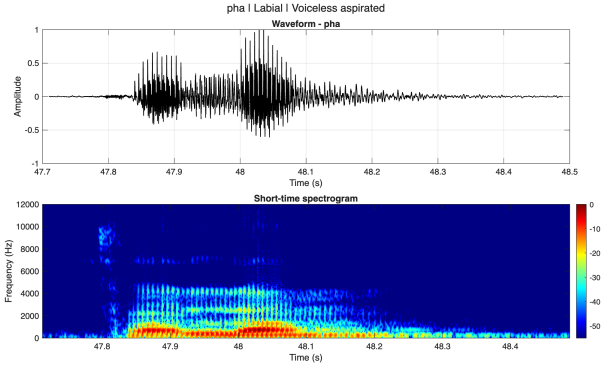
(e) na: nasal.

Figure 10: Waveforms and spectrograms of the five apico-dental tokens.

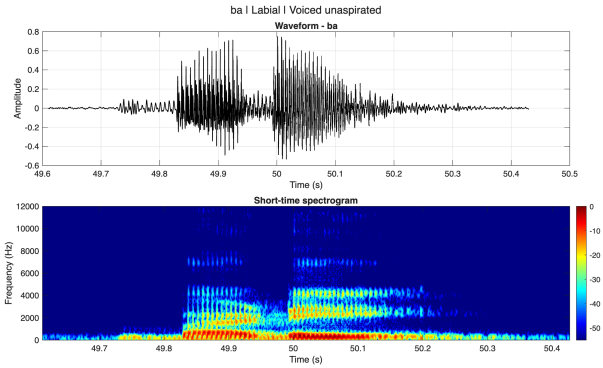
10.5 Labial Group



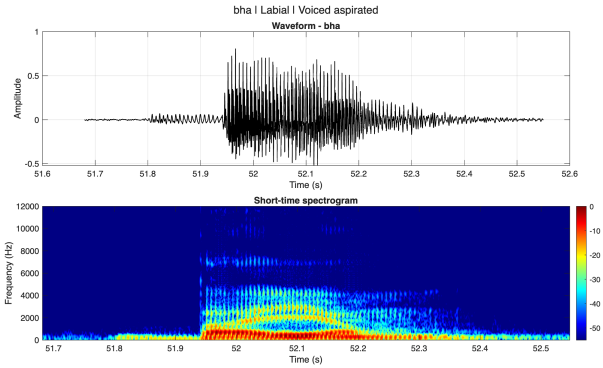
(a) pa: voiceless unaspirated.



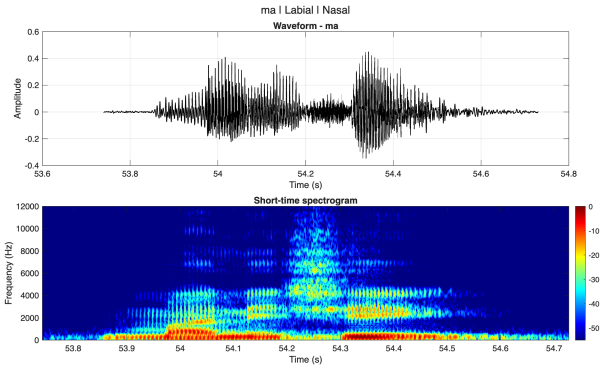
(b) pha: voiceless aspirated.



(c) ba: voiced unaspirated.



(d) bha: voiced aspirated.



(e) ma: nasal.

Figure 11: Waveforms and spectrograms of the five labial tokens.

Telugu consonants arranged by place and manner

```
VELAR
ka       kha       ga       gha       nga
Voiceless unaspirated | Voiceless aspirated | Voiced unaspirated | Voiced aspirated | Nasal

PALATAL
ca       cha       ja       jha       nya
Voiceless unaspirated | Voiceless aspirated | Voiced unaspirated | Voiced aspirated | Nasal

RETROFLEX
Ta       Tha       Da       Dha       Na
Voiceless unaspirated | Voiceless aspirated | Voiced unaspirated | Voiced aspirated | Nasal

APICO-DENTAL
ta       tha       da       dha       na
Voiceless unaspirated | Voiceless aspirated | Voiced unaspirated | Voiced aspirated | Nasal

LABIAL
pa       pha       ba       bha       ma
Voiceless unaspirated | Voiceless aspirated | Voiced unaspirated | Voiced aspirated | Nasal
```

Figure 12: Consonant classification arranged by place and manner of articulation.

11 Conclusion

Narrowband harmonic analysis, AMDF pitch estimation, and wideband LPC formant tracking were carried out for Amma (/a/), Illu (/i/), and Uduta (/u/). The harmonic-spacing and AMDF pitch estimates agreed within 1.11–1.74 Hz, giving pitch values close to 150–160 Hz. The formant estimates showed the expected broad vowel contrast: Illu had the lowest F1 and highest F2, while Uduta had the lowest F2.

The consonant recording contained 25 CV tokens organised in five articulation groups. Their waveform and spectrogram representations showed the closure, release, aspiration, voicing, nasal resonance, and vowel-transition cues used in consonant classification. Comparisons within each place group made the voicing and aspiration contrasts easier to interpret, while the burst spectrum and formant transitions provided qualitative evidence for place distinctions. The analysis therefore combines the articulatory organisation in the assignment chart with acoustic evidence from the recorded Telugu tokens.

References

[1] EE5334, *Assignment 1*, Objectives 1 and 2, course handout, 2026.

[2] P. Bhaskararao and A. Ray, "Telugu," *Journal of the International Phonetic Association*, vol. 47, no. 2, pp. 231–241, 2017. doi: 10.1017/S0025100316000207.

[3] J. Coleman, "Acoustic Structure of Consonants," University of Oxford, Phonetics teaching resource. Available: https://www.phon.ox.ac.uk/jcoleman/consonant_acoustics.htm.

[4] The Education University of Hong Kong, "Acoustic Aspects of Consonants," *Phonetics and Phonology* teaching resource. Available: https://corpus.eduhk.hk/english_pronunciation/index.php/3-2-acoustic-aspects-of-consonants/.